

LECTURE 03 · REVIEW NOTES

Risk and Return

How risk and return are measured for a single asset and for a portfolio, and why correlation is the foundation of Modern Portfolio Theory.

About these notes

These notes review the material of Lecture 03. Each concept is explained in full, and each numerical example is worked out step by step, with every intermediate figure shown. Interactive activities and student presentations completed during class are summarized in general terms and marked as in-class activities. Readings: Bodie, Kane and Marcus, Chapters 5.4 and 6.1.

1. Long and inverse index ETFs

An index ETF such as QQQ, which tracks the NASDAQ-100, rises and falls with the index. An inverse leveraged ETF such as SQQQ targets $-3\times$ the index's *daily* return, so it gains when the index falls and loses when it rises. On any single day the relationship is exact: if QQQ rises 2%, SQQQ falls about $3 \times 2\% = 6\%$.

Over longer periods the relationship breaks down. Because SQQQ resets its leverage every night, its return over a year is *not* -3 times QQQ's annual return, and the gap widens with volatility. This is the volatility decay discussed in Lecture 02. SQQQ is, in effect, a packaged bet against the index, which leads to the mechanism behind any such bet: the short sale.

IN-CLASS ACTIVITY · Student presentation: index ETFs vs. inverse ETFs

A student team compared \$10,000 invested in QQQ and in SQQQ through the 2022 bear market and the 2023 bull market. The comparison showed SQQQ gaining while the index fell in 2022 and losing most of its value as the index recovered in 2023, and the class discussed why the long-run result differs from $-3\times$ the index's total return.

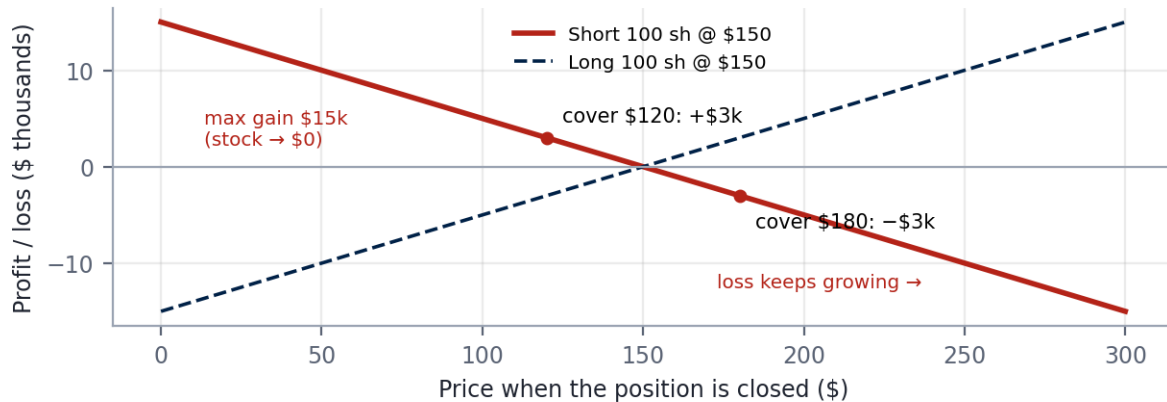
2. Short selling: the mechanism behind inverse funds

Short selling is the sale of shares the seller does not own. The shares are borrowed through a broker and sold at the current price; later they are bought back ("covered") and returned to the lender. If the price has fallen in the meantime, the short seller keeps the difference; if it has risen, the short seller pays it. Three groups use short sales: **speculators** who expect a price to fall, **hedge funds** that short one position to offset the risk of a long position, and **arbitrageurs** who short an overpriced security while buying a related underpriced one.

The process has five steps: borrow the shares, sell them at the market price P_0 , hold the position, cover by buying the shares back at P_1 , and return them. Per share, before costs:

$$\text{Short-sale profit per share} = P_0 - P_1$$

For instance, a short sale at \$150 covered at \$120 earns \$30 per share, while one covered at \$180 loses \$30 per share. The payoff is the mirror image of a long position, with one crucial difference shown in the figure below: the gain is capped, because a price cannot fall below zero, but the loss is not, because a price can keep rising.



Profit or loss at closing for 100 shares sold short at \$150, compared with a long position of the same size. Borrow fees and dividends are ignored.

This asymmetry is the key risk of shorting. A long position can lose at most 100% of the amount paid. A short position can lose more than 100% of the sale proceeds. In practice, a rising price also reduces the equity in the short seller's margin account, and a margin call can force the position closed before the seller would choose.

IN-CLASS ACTIVITY · Short-sale P&L calculator

Short-sale payoffs were explored in class with an interactive calculator for 100 shares sold short at \$150, varying the cover price to see the profit or loss per share, in total and as a return.

SQQQ: short selling, packaged

SQQQ delivers a short position on the whole NASDAQ-100 without the machinery of a short sale. Instead of borrowing shares of NVIDIA, Apple, Microsoft, Amazon, Tesla and the other 95 index members, the fund enters swap contracts that pay it the negative of the index's return. Investors buy SQQQ like any ETF, with no margin account, no borrow fee and no margin calls, and the maximum loss is the amount invested.

	Manual short sale	SQQQ (ProShares)
How it works	Borrow shares, sell, buy back later	Swap contracts on the NASDAQ-100
Payoff	$-1 \times$ the stock's return over the holding period	$-3 \times$ the index's <i>daily</i> return
Account needed	Margin account	Any brokerage account
Costs and risks	Borrow fees, dividends owed, margin calls, unlimited loss	Expense ratio and volatility decay; loss capped at amount invested
Beta / correlation with QQQ	n/a (single stock)	Beta -3.00 ; daily correlation -1.00

The beta of -3 and correlation of -1 describe the fund's *daily* behavior, which is exactly what it is built to deliver. They introduce the main subject of this topic: how the correlation between two assets determines the risk of holding them together.

3. Return and risk for a single asset

The **holding-period return** (HPR) measures what an investment earned over the time it was held. It has two sources, the change in price (the capital gain or loss) and any cash income received, such as dividends:

$$\text{HPR} = (P_1 - P_0 + D) \div P_0 = \text{capital-gain yield} + \text{dividend yield}$$

WORKED EXAMPLE 1

Holding-period return

A stock is bought at \$50, pays a \$1.50 dividend during the year, and is sold at \$54. (Illustrative numbers.)

Step	Calculation	Result
1	Capital gain = $P_1 - P_0 = \$54 - \50	\$4.00
2	Total dollar return = capital gain + dividend = $\$4.00 + \1.50	\$5.50
3	HPR = $\$5.50 \div \50	11.0%
4	Split: capital-gain yield = $\$4 \div \50 ; dividend yield = $\$1.50 \div \50	8.0% + 3.0% = 11.0%

A realized HPR is historical and observable. Investment decisions, however, rely on the **expected return** $E[r]$, the return anticipated before the outcome is known. It can be estimated from the average of past returns or from a model such as the CAPM.

Risk is uncertainty about that return. It is measured by the **standard deviation** σ , the typical size of the deviations of returns around their mean:

$$\sigma = \sqrt{E[(r_t - \bar{r})^2]}$$

A higher σ means outcomes are more dispersed and harder to predict. Because investors dislike that uncertainty, they require a higher expected return to hold riskier assets. The additional expected return over a risk-free rate is the **risk premium**.

WORKED EXAMPLE 2

Mean and standard deviation from five annual returns

A stock returned 12%, −8%, 20%, 5% and 16% over five years. (Illustrative numbers.)

Year	Return r_t	Deviation $r_t - \bar{r}$	Squared deviation
1	12%	$12 - 9 = 3$	9
2	−8%	$-8 - 9 = -17$	289
3	20%	$20 - 9 = 11$	121
4	5%	$5 - 9 = -4$	16
5	16%	$16 - 9 = 7$	49
Sum	45%	0 (always)	484

Step	Calculation	Result
1	Mean $\bar{r} = 45\% \div 5$	9.0%
2	Squared deviations (table above), summed	484 (%²)
3	Variance, dividing by $n = 5$ (the expectation definition above)	96.8 (%²)
4	$\sigma = \sqrt{96.8}$	9.84%
5	Sample version, dividing by $n - 1 = 4$: variance = 121, $\sigma = \sqrt{121}$	11.0%

Dividing by n treats the five years as the entire population of outcomes. When a sample of history is used to *estimate* an unknown future σ , statistical software and Excel's STDEV.S divide by $n - 1$, which produces a slightly larger number. With long histories the difference becomes negligible.

Key insight

Every asset is described by a pair of numbers, $(E[r], \sigma)$, never by return alone. Two assets can offer the same expected return with very different risk. The objective is the most expected return per unit of risk.

4. Modern Portfolio Theory

Modern Portfolio Theory (MPT), developed by Harry Markowitz, is a systematic method for finding the best portfolio for an investor. Its **inputs** are the expected return and standard deviation of each security, the correlation between every pair of securities, and the investor's tolerance for risk. The **process** computes the expected return and risk of every possible combination of weights, identifies the efficient frontier, and ranks portfolios by their risk-return trade-off. The **output** is the optimal portfolio for that investor: the highest return per unit of the risk the investor is willing to bear.

The central idea is that **a security should not be judged in isolation**. What matters is how it changes the risk of the whole portfolio. A volatile stock that tends to move opposite to the rest of the holdings can *reduce* total portfolio risk.

The two-asset formulas

For a portfolio with weight w_A in asset A and $w_B = 1 - w_A$ in asset B:

$$E[r_p] = w_A \cdot E[r_A] + w_B \cdot E[r_B]$$

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2 \cdot w_A \cdot w_B \cdot \sigma_A \cdot \sigma_B \cdot \rho_{AB}$$

The expected return is a simple weighted average, and **correlation plays no role in it**. The variance has three terms: each asset's own variance scaled by its squared weight, plus a cross-term. The cross-term contains the covariance, $\text{Cov}(A, B) = \sigma_A \sigma_B \rho_{AB}$, so the lower the correlation ρ , the smaller the portfolio variance. This is the entire source of the diversification benefit.

Correlation ρ	Meaning	Diversification benefit
+1	Perfect positive: the assets always move together	None. σ_p is the weighted average of the σ 's
0	Uncorrelated: movements are unrelated	Substantial
-1	Perfect negative: the assets always move in opposite directions	Maximum. Some mix has $\sigma_p = 0$

Real stocks typically have correlations of about 0.3 to 0.6 with one another: positive but well below 1. Combining them therefore always reduces risk relative to the weighted average, although it can never eliminate it.

WORKED EXAMPLE 3

Portfolio return and risk: 60% stocks, 40% bonds

Asset	E[r]	σ	Weight
A: Stocks	15%	30%	$w_A = 60\%$
B: Bonds	7%	12%	$w_B = 40\%$

The correlation is $\rho_{AB} = 0.20$. The goal is to find $E[r_p]$, σ_p^2 and σ_p . Working in decimals keeps the variance units consistent.

Step	Calculation	Result
1	Expected return = $0.60 \times 15\% + 0.40 \times 7\% = 9.0\% + 2.8\%$	$E[r_p] = 11.8\%$
2a	Stock term = $w_A^2 \sigma_A^2 = (0.6)^2 \times (0.30)^2 = 0.36 \times 0.09$	0.032400
2b	Bond term = $w_B^2 \sigma_B^2 = (0.4)^2 \times (0.12)^2 = 0.16 \times 0.0144$	0.002304
2c	Cross-term = $2 \times 0.6 \times 0.4 \times 0.30 \times 0.12 \times 0.20 = 0.48 \times 0.036 \times 0.20$	0.003456
2	Variance = $0.032400 + 0.002304 + 0.003456$	$\sigma_p^2 = 0.03816$
3	Standard deviation = $\sqrt{0.03816}$	$\sigma_p = 19.5\%$
4	Comparison: weighted-average $\sigma = 0.6 \times 30\% + 0.4 \times 12\% = 18\% + 4.8\%$	22.8%

The portfolio's risk (19.5%) is well below the weighted average of the two risks (22.8%), while its expected return is exactly the weighted average. The 3.3-point reduction in risk costs nothing in return; it comes entirely from a correlation below +1.

WORKED EXAMPLE 4**How much does correlation matter?**

The same assets and weights are kept, and only ρ changes. Terms 2a and 2b do not depend on ρ and always sum to 0.034704. The cross-term is $2 \times 0.6 \times 0.4 \times 0.30 \times 0.12 \times \rho = 0.01728 \times \rho$.

Correlation ρ	Cross-term $0.01728 \times \rho$	Variance σ_p^2	σ_p	$E[r_p]$
+1 (no benefit)	+0.017280	$0.034704 + 0.017280 = 0.051984$	22.8%	11.8%
+0.20	+0.003456	$0.034704 + 0.003456 = 0.038160$	19.5%	11.8%
0	0	0.034704	18.6%	11.8%
-1 (maximum)	-0.017280	$0.034704 - 0.017280 = 0.017424$	13.2%	11.8%

Two checks confirm the extremes. At $\rho = +1$ the variance is a perfect square, so $\sigma_p = 0.6 \times 30\% + 0.4 \times 12\% = 22.8\%$, the weighted average. At $\rho = -1$ it is also a perfect square, so $\sigma_p = |0.6 \times 30\% - 0.4 \times 12\%| = |18\% - 4.8\%| = 13.2\%$. As ρ falls from +1 to -1, risk drops from 22.8% to 13.2% while the expected return stays at 11.8%. The reward is unchanged and the risk is lower: the free lunch of diversification.

WORKED EXAMPLE 5**Two special portfolios: zero risk and minimum variance**

Zero risk when $\rho = -1$. With perfectly negative correlation the two risks can cancel exactly. Setting $\sigma_p = |w_A \sigma_A - (1 - w_A) \sigma_B| = 0$ and solving:

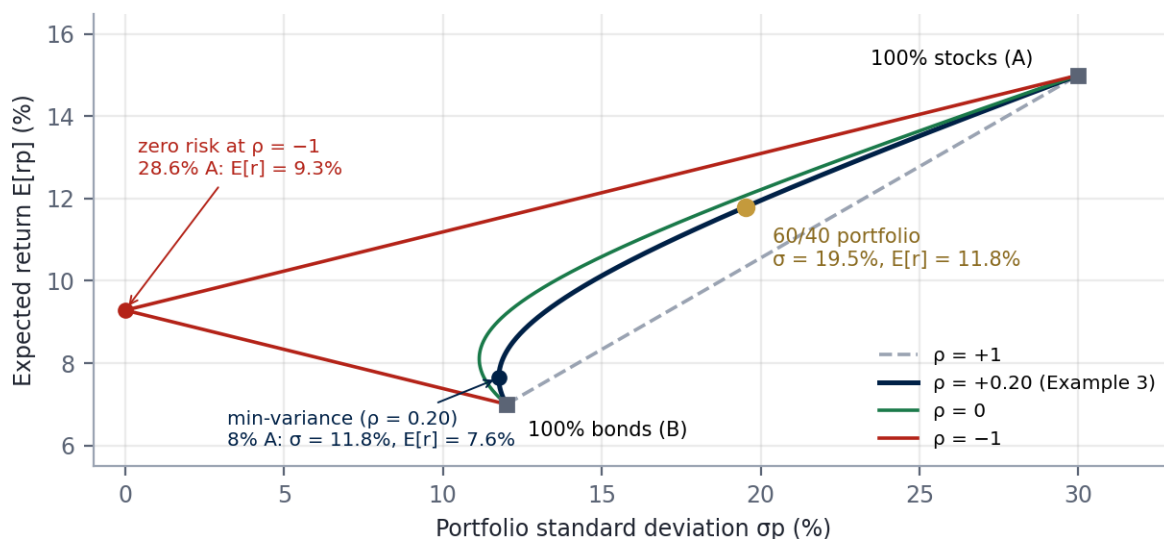
Step	Calculation	Result
1	$w_A \times 30\% = (1 - w_A) \times 12\%$	$30w_A = 12 - 12w_A$
2	$w_A = \sigma_B \div (\sigma_A + \sigma_B) = 12 \div 42$	$w_A = 28.6\%, w_B = 71.4\%$
3	$E[r_p] = 0.286 \times 15\% + 0.714 \times 7\%$	9.29% with $\sigma_p = 0$

Minimum variance when $\rho = 0.20$. With a realistic correlation, risk cannot reach zero, but one mix has the lowest possible risk. Its weight on A is:

$$w_A^* = (\sigma_B^2 - \sigma_A \sigma_B \rho) \div (\sigma_A^2 + \sigma_B^2 - 2\sigma_A \sigma_B \rho)$$

Step	Calculation	Result
1	Numerator = $0.0144 - (0.30)(0.12)(0.20) = 0.0144 - 0.0072$	0.0072
2	Denominator = $0.09 + 0.0144 - 2(0.0072) = 0.09$	0.0900
3	$w_A^* = 0.0072 \div 0.09$	8% stocks, 92% bonds
4	$E[r_p] = 0.08 \times 15\% + 0.92 \times 7\%$	7.64%
5	$\sigma_p^2 = 0.08^2(0.09) + 0.92^2(0.0144) + 2(0.08)(0.92)(0.0072) = 0.000576 + 0.012188 + 0.001060$	0.013824
6	$\sigma_p = \sqrt{0.013824}$	11.76%

An all-bond portfolio has $\sigma = 12\%$ and $E[r] = 7\%$. Moving just 8% of the money into the *riskier* asset lowers risk to 11.76% *and* raises the expected return to 7.64%. The all-bond portfolio is therefore **inefficient**: no investor who cares only about $E[r]$ and σ should hold it. Adding a volatile asset with low correlation makes the portfolio safer.



Every mix of stocks (A) and bonds (B) for four correlations. A lower ρ bends the curve further to the left: the same expected returns at lower risk. Only the upper part of each curve, from the minimum-variance point upward, is efficient.

5. The efficient frontier

For two assets, every possible portfolio lies on a single curve. The part of that curve running from the minimum-variance portfolio upward is the **efficient frontier**: for each level of risk, it offers the highest expected return. Portfolios on the lower part of the curve are inefficient, because another portfolio with the same σ offers more return.

With many assets, the possible portfolios fill an entire region rather than a single line, and individual assets sit to the right of the frontier: each can be matched by a diversified

portfolio with the same return and less risk. The efficient frontier is the **upper-left boundary** of that region. When correlations are lower, the whole frontier shifts to the left, improving the risk-return trade-off available to every investor.

IN-CLASS ACTIVITY · Correlation and the efficient frontier

Two interactive tools were used in class. The first varied the portfolio weight and the correlation for the stock-bond pair above, showing that $E[r_p]$ does not change with ρ while σ_p falls. The second built the frontier step by step from nine asset classes, from Treasury bonds to emerging-market equities, first plotting the individual assets, then the portfolios formed from them, and finally the efficient boundary.

6. Key takeaways

Return and risk come as a pair. A single asset's return is its capital gain plus income, and its risk is the standard deviation σ of that return. Investors require a risk premium for bearing higher σ , so assets are compared on $(E[r], \sigma)$, not on return alone.

Portfolio return is an average; portfolio risk is not. $E[r_p]$ is always the weighted average of the assets' expected returns and does not depend on correlation. σ_p^2 contains the cross-term $2w_A w_B \sigma_A \sigma_B \rho$, so portfolio risk falls as correlation falls.

Correlation drives the diversification benefit. At $\rho = +1$ there is no benefit. Below $+1$ the portfolio is less risky than the weighted average, and at $\rho = -1$ risk can be eliminated entirely. Typical stock correlations of 0.3 to 0.6 mean diversification always helps.

The efficient frontier is where rational investors choose. It is the upper-left boundary of all possible portfolios: the maximum expected return for each level of risk. Lower correlations shift it to the left.

The free lunch of investing

Combining assets with low or negative correlation reduces portfolio risk without reducing expected return. Changing ρ leaves $E[r_p]$ unchanged but shrinks σ_p . This asymmetry is the central message of Modern Portfolio Theory. The next topic, asset allocation, combines the risky portfolio with a risk-free asset through the Capital Allocation Line (CAL) to find the optimal risky portfolio.

Examples 3 to 5 use the stock and bond figures presented in Lecture 03; Examples 1 and 2 use illustrative numbers.